EV Drive Clean Rebates & Market Dynamics

Comprehensive Python Data Analysis | 2017–2026

PROJECT REPORT SUMMARY

An analytical review of EV rebate adoption, consumer powertrain preferences, financial incentives, geographic distribution, seasonality, and estimated environmental benefits.

Dataset	EV Drive Clean Rebate transactions
Source	Official Government Open Data Portal / NYSERDA
Coverage	2017–2026
Initial records	239,278
Final analytical records	225,428
Manufacturers	34
Vehicle models	148
Counties	62

Prepared from the supplied Google Colab notebook. Findings and terminology are based on the analysis contained in the notebook.

1. Executive Summary

This project analyzes the EV Drive Clean Rebate administrative dataset to understand how electric-vehicle rebate participation evolved from 2017 to 2026. The analysis combines data quality preparation, exploratory data analysis, feature engineering, univariate, bivariate and multivariate analysis, and statistical interpretation to evaluate adoption patterns and estimated environmental outcomes.

After cleaning and validation, 225,428 unique records were retained for analysis. The final dataset contains vehicle manufacturer and model information, location, EV technology type, transaction type, rebate amount, estimated annual GHG reductions, and estimated annual petroleum reductions.

The analysis shows a strong BEV preference, a large concentration of transactions around the $500 incentive, substantial geographic concentration, recurring seasonal variation, and a strong relationship between petroleum and GHG reduction measures. The notebook also introduces efficiency indicators to compare environmental benefit against rebate spending.

Key KPI	Meaning
225,428	Final unique records
66.8%	BEV share
$885.01	Mean rebate
$500	Median & mode rebate
518.51 gal	Average annual petroleum reduction
2.44 MT CO_2e	Average annual GHG reduction
186,353	Medium-impact records
39,255	Nassau County records

Portfolio-level takeaway: the program is dominated by BEVs and lower-value rebates, while environmental outcomes are concentrated within the Medium Impact tier. The notebook interprets these patterns as evidence for more targeted and efficiency-aware policy design.

2. Data Preparation & Quality Management

Initial Dataset Profile

• The raw dataset contained 239,278 rows and 11 columns.

• Seven missing values were identified across County, ZIP, and Transaction Type.

• 13,845 exact duplicate rows were identified using all available columns.

• One invalid rebate record and four negative petroleum-reduction records were removed.

• After duplicate removal, 225,428 unique records were preserved for analysis.

• The final audit reported zero remaining missing values and 1,644 unique ZIP identifiers.

Cleaning & Transformation Approach

• Standardized column names to lowercase, underscore-separated headers.

• Replaced unresolved missing County, ZIP, and Transaction Type values with 'Unknown' to preserve records.

• Converted rebate amounts, petroleum reductions and GHG reductions into numeric formats.

• Converted Submitted Date into datetime format and extracted year/month fields.

• Normalized categorical text and unified selected manufacturer-name variations.

• Engineered GHG-reduction-per-dollar and petroleum-reduction-per-dollar efficiency indicators.

• Created Environmental Impact and Rebate Value tiers for categorical comparison.

Feature Engineering

Engineered feature	Purpose
ghg_reduction_per_dollar	Annual GHG reduction relative to rebate amount
petroleum_reduction_per_dollar	Annual petroleum reduction relative to rebate amount
environmental_impact_tier	Low / Medium / High environmental grouping
rebate_tier	Low / Mid / High incentive grouping
rebate_year / rebate_month	Time-based trend and seasonality analysis

3. EV Market Dynamics & Adoption Patterns

Yearly Adoption Trend

The time-series analysis shows rebate activity rising from a low baseline in 2017, accelerating strongly through the middle of the period, and then stabilizing before the 2026 endpoint. The project interprets this trajectory as evidence of substantial growth in EV participation.

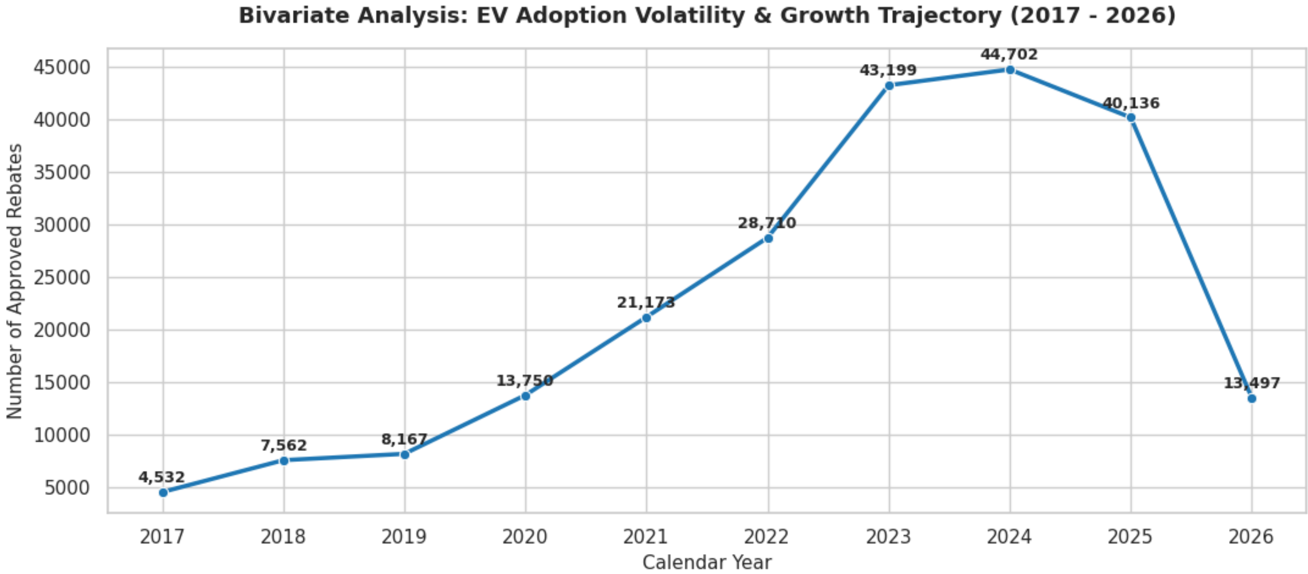

Figure 1. EV adoption volatility and growth trajectory.

Powertrain Preference

BEVs account for 150,664 records (66.8%), while PHEVs account for 74,764 records (33.2%). This establishes a clear two-to-one preference for battery-electric powertrains within the analyzed records.

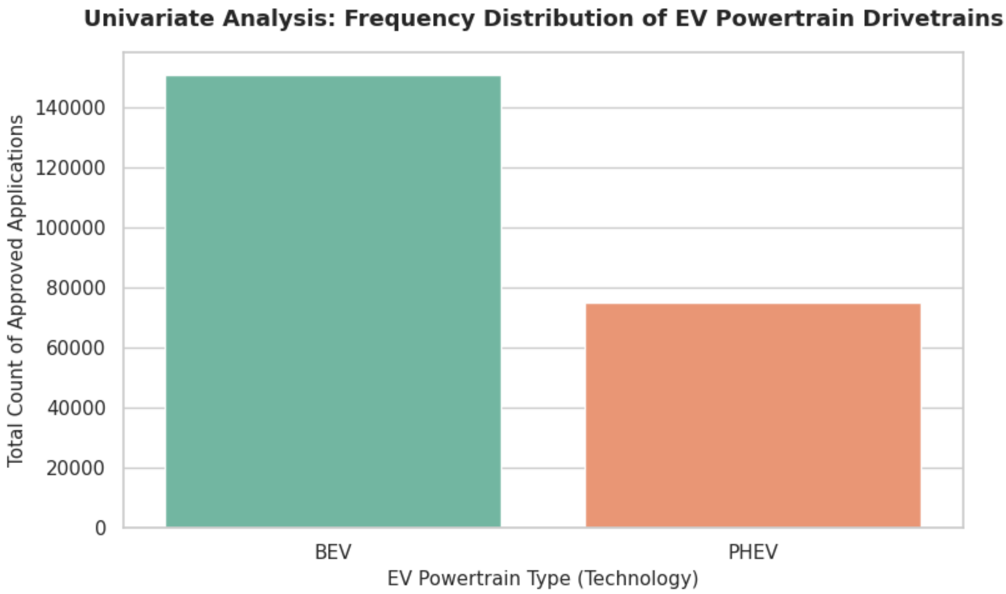

Figure 2. Distribution of EV powertrain types.

Rebate Distribution

The mean rebate is $885.01, while both the median and mode are $500. The engineered rebate tiers show 150,302 low-incentive records (≤$500), 25,480 mid-incentive records ($501–$1,100), and 49,646 high-incentive records (>$1,100).

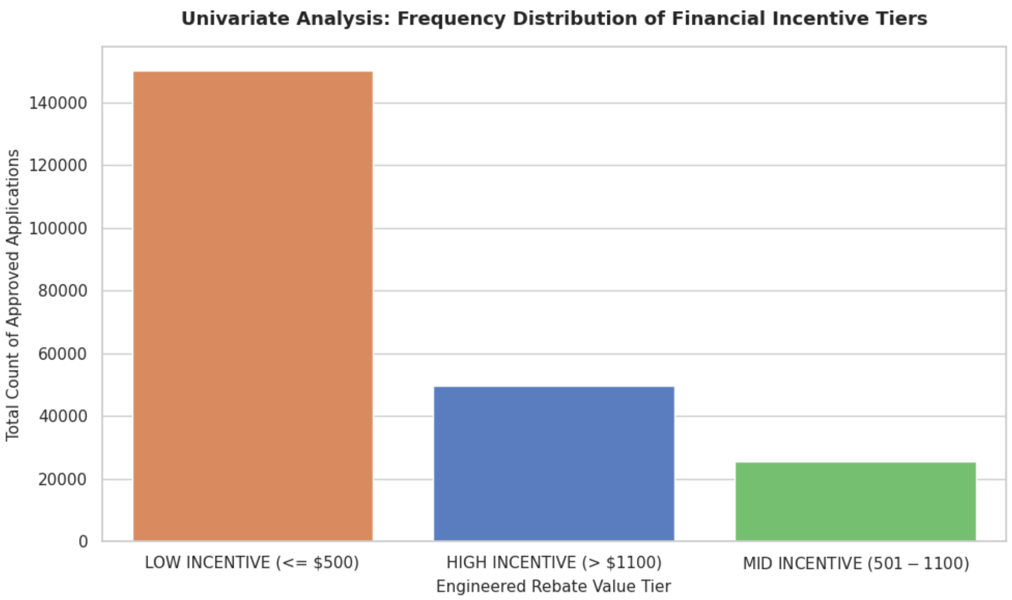

Figure 3. Frequency distribution of financial incentive tiers.

4. Geographic & Seasonal Patterns

Geographic Distribution

The dataset contains 62 counties and 1,644 unique ZIP identifiers after cleaning. Nassau County records are the highest at 39,255. The notebook identifies substantial geographic concentration and interprets adoption as uneven across regions.

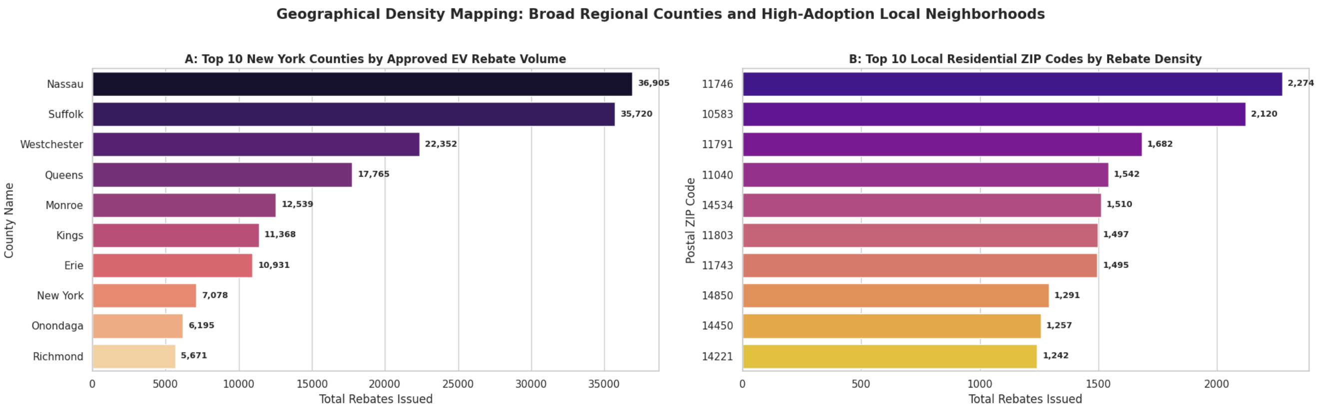

Figure 4. County and ZIP-level concentration of rebate transactions.

Seasonality

Monthly analysis shows recurring lower transaction volumes during January and February, followed by stronger activity through spring and summer. The notebook interprets this as a predictable seasonal pattern that can help anticipate administrative workload and program demand.

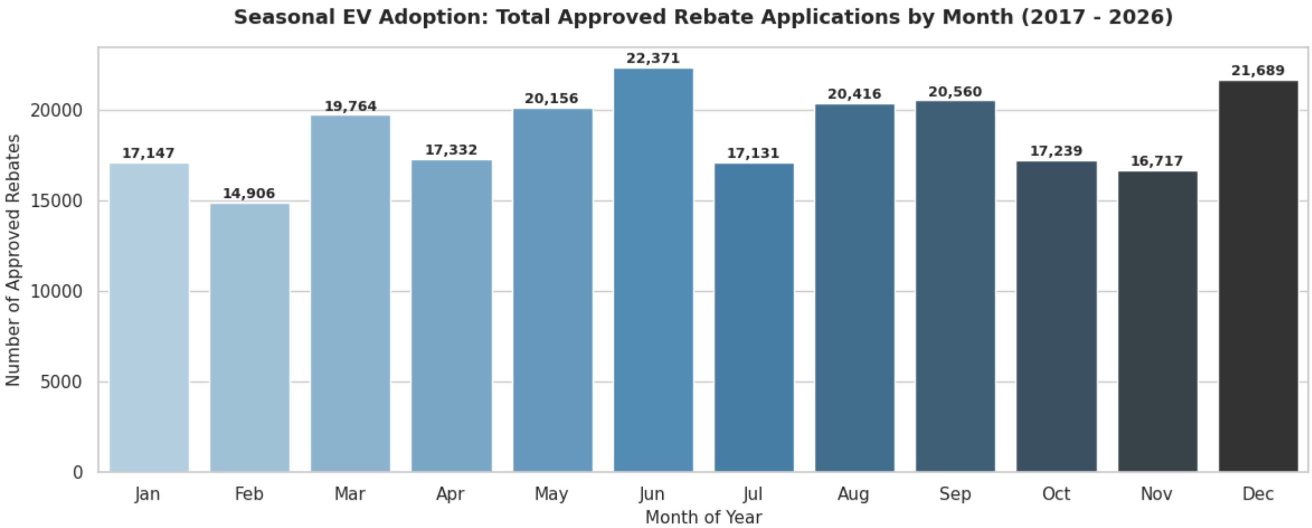

Figure 5. Seasonal EV adoption pattern by month.

5. Environmental Benefits & Efficiency Analysis

Environmental Impact Classification

The engineered environmental-impact classification places 186,353 records (82.7%) in the Medium Impact tier, 39,050 in Low Impact, and 25 in High Impact. The analyzed transactions are therefore strongly concentrated within the defined Medium Impact band.

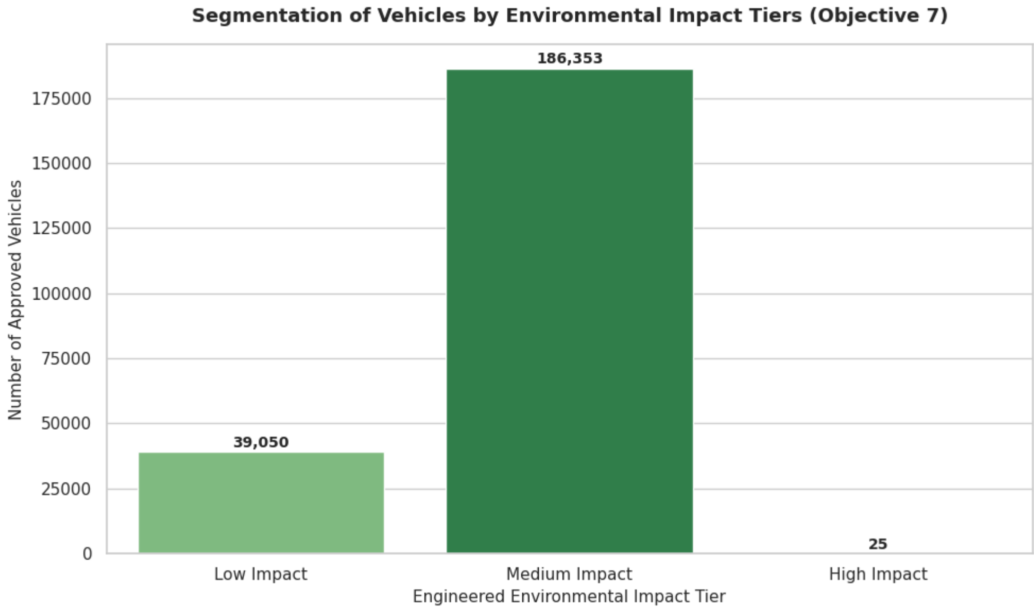

Figure 6. Distribution of engineered environmental impact tiers.

Environmental Performance Indicators

• Average annual petroleum reduction: 518.51 gallons per vehicle.

• Average annual GHG reduction: 2.44 metric tons CO_2e per vehicle.

• Petroleum-reduction efficiency averaged 0.793 gallons saved per rebate dollar.

• The notebook reports a strong relationship between GHG reduction and petroleum reduction.

• The correlation matrix reports 1.00 correlation between GHG and petroleum reductions, while rebate amount has a much weaker relationship with environmental measures.

Brand-Level Efficiency

The top-five-brand comparison shows noticeable differences in petroleum-savings efficiency per rebate dollar. The notebook uses this indicator to demonstrate that similar rebate structures can produce different environmental returns depending on vehicle mix.

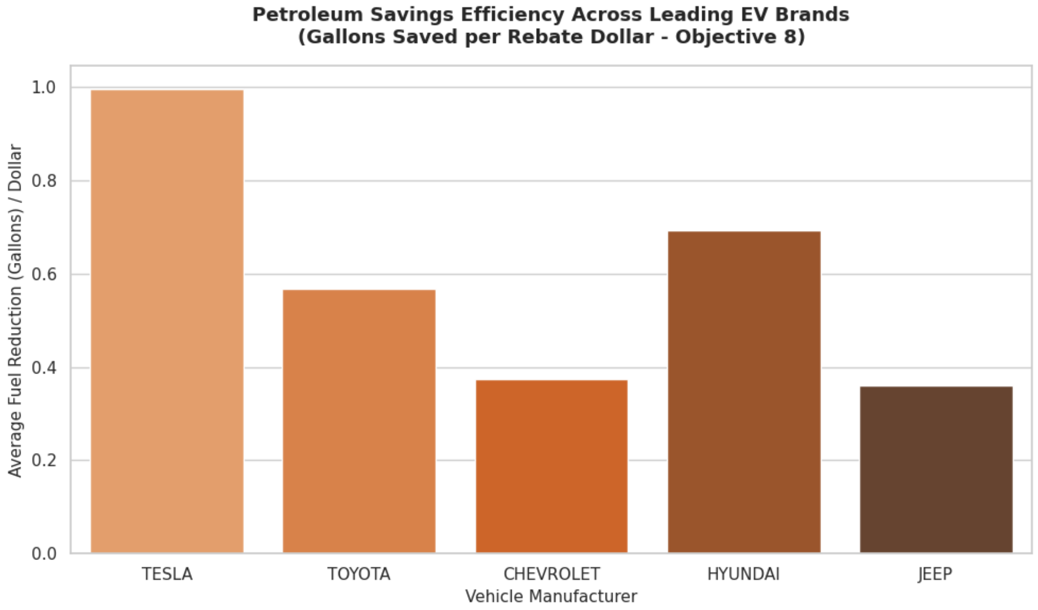

Figure 7. Petroleum savings efficiency across leading EV brands.

6. Statistical Interpretation & Strategic Recommendations

Distributional Findings

• Rebate amounts are positively skewed (reported skewness: 1.1299), reflecting a long right tail created by higher-value incentives.

• Petroleum reductions show strong negative skewness (−1.7384), while GHG reductions show negative skewness (−1.9685).

• GHG reduction has reported positive kurtosis (3.0509), indicating a heavier-tailed distribution for that measure.

• The median and mode of rebate amount are both $500, reinforcing the dominance of the base incentive level.

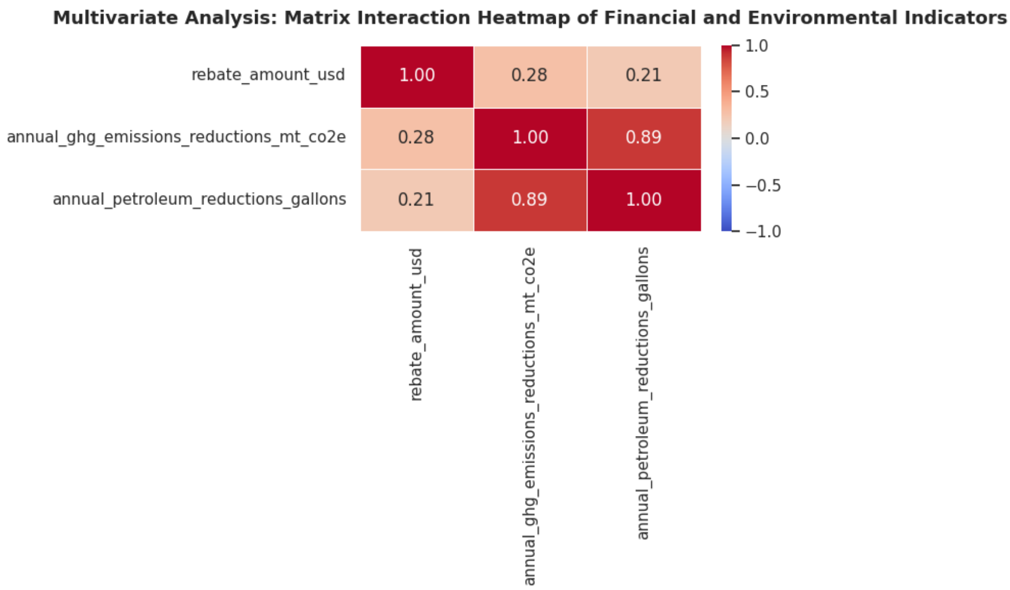

Figure 8. Correlation matrix for financial and environmental indicators.

Strategic Recommendations from the Notebook

1. Efficiency-linked subsidy design — Consider a tiered incentive model that links rebate value more directly to measurable vehicle efficiency and environmental outcomes.

2. Reduce concentration risk — The notebook identifies strong concentration around Tesla and the Model Y; a diversified incentive strategy could support broader manufacturer participation.

3. Target geographic gaps — Use county and ZIP-level adoption patterns to prioritize charging infrastructure and outreach in lower-adoption areas.

4. Plan operations around seasonality — Scale administrative and dealership-support capacity ahead of recurring spring/summer demand peaks.

7. Conclusion

The analysis of 225,428 cleaned and unique EV rebate transactions demonstrates a mature and highly structured rebate ecosystem. BEVs represent 66.8% of analyzed records, the $500 incentive is the dominant rebate level, and 82.7% of records fall within the engineered Medium Environmental Impact tier. The average estimated annual benefit is 518.51 gallons of petroleum reduction and 2.44 MT CO_2e of GHG reduction per vehicle.

From a data-analytics perspective, the project demonstrates an end-to-end Python workflow: data inspection, quality validation, cleaning, transformation, feature engineering, exploratory analysis, statistical interpretation, visualization, and business recommendations. The results provide a practical framework for evaluating EV incentive program participation and environmental efficiency.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab
Domain: Green Energy Transportation Analytics
Source: Official Government Open Data Portal / NYSERDA